

# Problem Set 6

POLS W4732

Fall 2026

Due November 20 before the TA section

**Artificial-intelligence policy.** Students may not use generative-AI systems (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, and collaboration within the registered problem-set group are permitted. **Every group member is responsible for understanding and being able to explain the entire submission.**

**Exercise 1** (Media bias and career concerns). *An incumbent's first-period performance is*

$$\pi_1 = \theta_I + e_1 + \varepsilon_1,$$

where  $\theta_I \sim N(m, \sigma_\theta^2)$  and effort costs  $e_1^2/2$ . Reelection is worth  $B$ . The voter compares the incumbent's posterior expected ability with a challenger of expected ability  $m$  and correctly anticipates equilibrium effort  $e_1^a$ .

1. *Environment A:  $\varepsilon_1 \sim N(b, \sigma_\varepsilon^2)$  and  $b$  is common knowledge. Derive the posterior mean, reelection cutoff, winning probability, and effort first-order condition.*
2. *Environment B: the true bias is  $b$ , but the voter believes it is  $\hat{b} \neq b$ . Repeat the analysis.*
3. *Environment C: the mean bias is known but its realization contains an additional independent normal component with variance  $\sigma_b^2$ . Repeat the analysis.*
4. *State the exact condition under which additive bias cancels from equilibrium effort. Evaluate the claim that favorable coverage must reduce effort.*

*Throughout the updating calculation the observed first-period shock is  $\varepsilon_1$  (not  $\varepsilon_2$ ).*

**Exercise 2** (Electoral advantage and information precision). *In the baseline two-period career-concern model, the voter obtains an additional payoff  $\kappa$  when the incumbent is reelected. Ability is  $N(m, \sigma_\theta^2)$ , performance noise is  $N(0, \sigma_\varepsilon^2)$ , office benefit is  $B$ , and effort cost is  $c(e)$ .*

1. *Derive the voter's posterior and show that the reelection cutoff is*

$$\pi^* = e_1^a + m - \frac{\kappa}{\lambda}, \quad \lambda = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\varepsilon^2}.$$

2. *Derive the equilibrium effort condition and state the second-order qualification.*
3. *For  $c(e) = e^2/2$ , test the assertion that more precise information always raises effort. Derive the sign rather than relying on verbal intuition.*
4. *Explain separately the accountability and electoral-security effects. Give one explicit modification of the baseline assumptions under which the precision comparative static could reverse, and state the condition needed.*